

Package #170

Embed a thin perforated stainless steel mesh sleeve (pore size ≥ 4 mm) within the catheter floor lumen, secured to the catheter wall,

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose drain conductance, and can a wider floor lumen restore flow? (plausibility bounds)". [fda_maude] 100 records retrieved live (1997-2032). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000032: Malfunction [PUMP MMT-1780KPK 670G PATHWAY BLACK MG]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: '<i>E. coli</i> bacterium near corrugated surfaces: near-surface swimming, escape, and hydrodynamic trapping.'; 'Computational Fluid Dynamics Reveals Mass Transfer Limitations in a Pilot-Scale Microbial Electrolysis Cell.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Perforated metal liner (mesh) used in chemical reactor tubes to prevent fouling while allowing fluid flow through the perforations, maintaining conduit integrity under pressure.

Contents:

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Status: ENGINEERING_DEFINITION. Not physically validated.